

O-RAN Near-Real-Time RIC (xApp Container)

**5G NR gNB
& UE Mobility**

E2SM-KPM

RSRP Sequence
 R_t in R^{50}

**Pillar 1: Kinematic
Physics Layer**

$v_t = dr/dt$
 $a_t = d^2r/dt^2$
Energies (E_v, E_a)
→ X_{kin} in R^{64}

**Pillar 3: TinyML
Quantized Engine**

INT8 MLP (56.8% RAM v)
Pruned DT (0.096 ms)
→ Prob. p_{hat}

**Pillar 2: DAT
Cost Engine**

$C(P_{th}) = FP + 5*FN$
Opt. $P_{th}^* = 0.35$
81.9% RLF Drop v
→ HO Decision